Attiq Ur Rehman

WIRELESS COMMUNICATIONS / 6G NTN / AI FOR NETWORKS

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Research Profile

Electrical and computer engineering researcher focused on 6G terrestrial and non-terrestrial network integration. Research spans cross-layer resource allocation using multi-agent reinforcement learning, interference management, and deep learning for Doppler compensation in very low Earth orbit satellite links. Experience in wireless system simulation, collaborative research, and project leadership.

Education

Master's in Electrical and Computer Engineering Mar. 2024 – Feb. 2026
Inha University Incheon, Republic of Korea

GPA: 4.13/4.50 | **Advisor:** KyungHi Chang
Thesis: Cross-Layer Multi-Agent PPO Resource Allocation in 6G A2G-TN Systems and Dual-Head Temporal Convolutional Network (TCN) Doppler Compensation for VLEO Satellites.

Bachelor's in Electrical Engineering Dec. 2016 – Nov. 2020
University of Sargodha Sargodha, Pakistan

GPA: 3.45/4.00 | **Advisor:** Engr. Nauman Ahmad Tariq
Thesis: IoT-Based Automatic Solar Panel Cleaning System, using Arduino UNO and NodeMCU ESP8266.

Research Experience

Research Assistant / Project Leader Mar. 2024 – Feb. 2026
Mobile Telecommunication Research Lab (MTRL) Link Incheon, Republic of Korea

- Designed and implemented a 3D wireless transmission simulator for supersonic mobile communications to support the development and evaluation of high-throughput 6G air-to-ground links.
- Investigated cross-layer resource allocation using multi-agent proximal policy optimization (PPO) for integrated air-to-ground and terrestrial networks.
- Researched dual-head temporal convolutional networks for wide-range Doppler acquisition and cyclic-prefix-aided tracking in very low Earth orbit (VLEO) satellite links.
- Led development of the laboratory's 3D communication framework and extended the research toward integrated LEO satellite and terrestrial networks, focusing on coverage and capacity.
- Mentored research colleagues in network topology selection, link-budget analysis, system-model validation, and simulation-based performance evaluation for integrated terrestrial and non-terrestrial networks.

Technical Skills

Programming: Python (advanced); MATLAB (intermediate).
Methods: Multi-agent PPO; reinforcement learning; temporal convolutional networks; wireless system modelling; resource allocation; Doppler compensation.
Simulation tools: Vienna (advanced); MATLAB Satellite Communications, Phased Array System, Antenna, 5G (3GPP NR link-level), and Reinforcement Learning toolboxes.
Visual communication: Adobe Illustrator and Adobe Photoshop (advanced).

Journal Publications

[1] **A. U. Rehman**, S. Sualiheen, and K. H. Chang, "Coarse-to-fine Doppler compensation in 6G VLEO: Dual-head temporal convolutional network acquisition with CP-aided tracking," *IEEE Transactions on Communications*, vol. 74, pp. 7539–7555, 2026. Link

[2] **A. U. Rehman**, S. Sualiheen, and K. H. Chang, "Adaptive resource allocation in 6G A2G-TN integrated system: A cross-layer multi-agent PPO approach," *Computer Network*, vol. 272, Art. no. 111655, Nov. 2025. Link

[3] S. Sualiheen, **A. U. Rehman**, and K. H. Chang, "6G air-to-ground connectivity: A codebook-based framework for interference management," *IEEE Journal on Selected Areas in Communications*, vol. 44, pp. 2687–2700, 2026. Link

[4] C. J. Pawase, **A. U. Rehman**, and K. H. Chang, "Enhanced 6G non-terrestrial network link performance using deep learning-based channel estimation and Doppler compensation techniques," *Journal of Communications and Networks(JCN)*, vol. 28, no. 1, pp. 73–85, Feb. 2026. Link

[5] A. Samim, **A. U. Rehman**, and K. H. Chang, "Intelligent handover management for 6G LEO satellite constellations: A predictive multi-agent PPO approach," *IEEE Transactions on Network and Service Management*, **accepted for publication**, 2026.

Manuscripts Under Review

[5] D. Khan, S. Aslam, **A. U. Rehman**, A. Afridi, A. Manan, and R. Wichman, "Context-aware multi-link communication for cooperative vehicle perception in 5G NR V2X networks," submitted to *IEEE Open J. Commun. Soc.*, **under review**.

[7] S. Sualiheen, **A. U. Rehman**, and K. H. Chang, "LEO satellites integration with terrestrial networks: An interference and communication-efficient price-based federated learning approach," submitted to *IEEE Transactions on Aerospace and Electronic Systems*, Nov. 2025, **under review**.

Magazine Article

[8] S. Sualiheen, **A. U. Rehman**, A. Samim, and K. H. Chang, "Trends in AI-based 6G non-terrestrial network technologies" (English translation; in Korean), *J-KICS Rev. Inf. Commun*, vol. 42, no. 6, pp. 3–12, Jun. 2025. Link

Conference Presentations

[9] **A. U. Rehman**, S. Aslam, and S. Y. Shin, "RSMA-ISAC power allocation for beam-hopping LEO satellite communications with distributed beam-local tracking," in *Proc. 5th International Conference on Mobile, Military, and Maritime IT Convergence (ICMIC 2026)*, Taipei, Taiwan, Aug. 2026. Link

[10] A. Samim, **A. U. Rehman**, S. Sualiheen, and K. H. Chang, "AI at the core of 6G: Integrating multi-orbit LEO satellites and terrestrial networks," in *Proc. KICS Winter Conf.*, Republic of Korea, Feb. 2025, pp. 712–713. Link

[11] **A. U. Rehman**, S. Sualiheen, and K. H. Chang, "Dynamic spectrum coordination mechanism for spectrum sharing between 6G NTN and TN," in *Proc. KICS Summer Conf.*, Republic of Korea, Jun. 2024, pp. 1835–1836. Link

[12] **A. U. Rehman**, S. Aslam, and S. Y. Shin, "Mitigating beam squint and sensing collapse in wideband ISAC LEO systems via power-split beamforming," in *Proc. KICS Summer Conference*, Republic of Korea, 2026. **Presented by Attiq Ur Rehman**. Link

Patent Applications

Republic of Korea | Inventors: **Attiq Ur Rehman** and KyungHi Chang

P1 Apparatus and Method for Carrier Frequency Offset Estimation Based on Dual-Head Temporal Convolutional Network for Wide-Range Doppler Compensation of Very Low Earth Orbit Satellite in Wireless Communication System.
Application no. **10-2026-0010135** | Filed Jan. 15, 2026.

P2 Adaptive Resource Allocation Through Cross-Layer Multi-Agent Proximal Policy Optimization in 6G A2G-TN Integrated System.
Application no. **10-2025-0041998** | Filed Apr. 1, 2025.

Software Copyright Registrations

Korea Copyright Commission | **Attiq Ur Rehman** and KyungHi Chang

- **Dual-Head TCN-Based Coarse-to-Fine Doppler Compensation for 6G VLEO Satellite Environments.** C-2026-008789 | Feb. 18, 2026.
- **Adaptive Resource Allocation in 6G Air-to-Ground Integrated Systems: A Cross-Layer Multi-Agent PPO Approach.** C-2024-050516 | Feb. 28, 2025.

Awards and Scholarships

- **Best Workshop Paper Award**, ICMIC 2026, Taipei, Taiwan — Aug. 2026
- **Jungseok International**, Inha University — 2024–2026
- **BK21 Scholarship**, Korean government-funded scholarship — 2024–2026
- **Project-Funded Scholarship**, MTRL — 2024–2026
- **Undergraduate Silver Medal**, University of Sargodha; CGPA 3.45/4.00.

Memberships & Professional Activities

- **Member, KICS** — Jun. 2024 – Present
  Korean Institute of Communications and Information Sciences.
- **Registered Engineer, Pakistan Engineering Council (PEC)**
  Registration no. **ELECT/87123**; registered Dec. 31, 2020.

Academic Engagement

Academic Seminar — Feb. 2025
Presented 6G Terrestrial–Non-Terrestrial Integration Systems: Challenges and Future Research Directions, covering hybrid network architectures, air-to-ground and LEO satellite links, and open research challenges.

Mobile Korea 2025: Regional Workshop on AI-Empowered Networks
Ministry of Science and ICT (MSIT) — Seoul, South Korea | Oct. 2025
Focus: AI-driven 6G network evolution, regional collaboration, and satellite communication integration.

Future Mobility Technology Workshop for the 6G Era
Korean Institute of Communications and Information Sciences (KICS) — Sep. 2024

Academic References

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